

Ryuji Hirayama

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UKRI Fellow, UKRI Frontier Research Guarantee Grant for ERC StG
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Research Interests

Volumetric display, Holography, High-performance computing, Acoustic levitation, Multimodal system, Haptics, Parametric audio, Human-computer interaction, 3D Ambigram, Computational fabrication

Education

03/2014–04/2017	PhD in Engineering Graduate School of Engineering, Chiba University, Japan Theme: Volumetric display containing multiple 2D images
03/2012.–04/2014	Master of Engineering Graduate School of Engineering, Chiba University, Japan
03/2008–04/2012	Bachelor of Engineering Faculty of Engineering, Chiba University, Japan

Experience

10/2023–present	Co-Founder/Hardware Lead AcoustoFab LTD, UK
08/2022–present	Lecturer Department of Computer Science, University College London, UK
07/2020–06/2022	Postdoctoral Research Fellow Department of Computer Science, University College London, UK
04/2019–06/2020	Postdoctoral Research Fellow Interact Lab, School of Engineering and Informatics, University of Sussex, UK
01/2019–03/2019	Rutherford Research Fellow Interact Lab, School of Engineering and Informatics, University of Sussex, UK
04/2018–12/2018	Research Fellow of the Japan Society for the Promotion of Science (PD) Masuda Lab, Faculty of Industrial Science and Technology, Tokyo University of Science, Japan
04/2018–09/2018	Visiting Research Fellow Interact Lab, School of Engineering and Informatics, University of Sussex, UK
04/2017–03/2018	Research Fellow of the Japan Society for the Promotion of Science (PD) Itot Lab, Graduate School of Engineering, Chiba University, Japan
04/2015–03/2017	Research Fellow of the Japan Society for the Promotion of Science (DC2) Itot Lab, Graduate School of Engineering, Chiba University, Japan
11/2014–03/2015	Research Assistant of the ImPACT Program Graduate School of Engineering, Chiba University, Japan
10/2014–03/2015	Teaching Assistant (Experiment of electrical and electronics engineering III) Faculty of Engineering, Chiba University
07/2013–10/2014	Student Assistant (Leaning support for undergraduates) Academic Link Centre, Chiba University

Journal Papers

1. J. Hardwick, Z. Morgan, and **R. Hirayama**, ‘Acoustophoretic System for Seed Separation Conveyor Belts,’ Nature Communications (accepted).
2. N. Kemsaram, J. Hardwick, J. Wang, B. Gautam, C. Besevli, G. Christopoulos, S. Dogra, L. Gao, A. Delibasi, D. M. Plasencia, O. Georgiou, M. Obrist, **R. Hirayama**, and S. Subramanian, ‘AcoustoBots: A swarm of robots for acoustophoretic multimodal interactions’, Frontiers in Robotics and AI 12, 1537101 (2025).
3. A. Achilleos, J. Hardwick, **R. Hirayama**, and S. Subramanian, ‘Omnidirectional Transmissive Acoustic Metasurfaces Based on Goldberg Polyhedra’, Advanced Functional Materials 2502899 (2025).
4. T. Dalsgaard, A. Bhatia, L. Gao, **R. Hirayama**, S. Subramanian, J. Bergström, and K. Hornbæk, ‘Ultrasound can deliver chemical stimulants to the skin and modulate their perception’, Scientific Reports 15, 10297 (2025).

5. H. Chen, S. Bansal, D. M. Plasencia, L. Di-Silvio, J. Huang, S. Subramanian, and **R. Hirayama**, ‘Omnidirectional and Multi-Material In Situ 3D Printing Using Acoustic Levitation’, *Advanced Materials Technologies* 2401792 (2024) featured on the back cover.
6. H. Chen, Y. Liu, S. Balabani, **R. Hirayama**, and J. Huang, ‘Machine Learning in Predicting Printable Biomaterial Formulations for Direct Ink Writing’, *Research* 6 (2023).
7. R. Montano, **R. Hirayama**, and D. M. Plasencia, ‘OpenMPD: A low-level presentation engine for Multimodal Particle-based Displays’, *ACM Transactions on Graphics* (accepted).
8. **R. Hirayama**, G. Christopoulos, D. M. Plasencia, and S. Subramanian, ‘High-speed acoustic holography with arbitrary scattering objects’, *Science Advances* 8, eabn7614 (2022).
9. D. M. Plasencia, **R. Hirayama**, R. Montano, and S. Subramanian, ‘GS-PAT: High-speed multi-point sound-fields for phased arrays of transducers’, *ACM Transactions on Graphics* 39, 138 (2020).
10. **R. Hirayama**, D. M. Plasencia, N. Masuda, and S. Subramanian, ‘A volumetric display for visual, tactile and audio presentation using acoustic trapping’, *Nature* 575, 320–323 (2019), featured in *Nature*.
11. D. Matsumoto, **R. Hirayama**, N. Hoshikawa, H. Nakayama, T. Shimobaba, T. Ito, and A. Shiraki, ‘Interactive directional volumetric display that keeps displaying directional image only to a particular person in real-time’, *OSA Continuum* 2(11), 3309–3322 (2019).
12. **R. Hirayama**, H. Nakayama, A. Shiraki, T. Kakue, T. Shimobaba, and T. Ito, ‘Projection of multiple directional images on a volume structure with refractive surfaces’, *Optics Express* 27(20), 27637–27648 (2019).
13. Y. Yamamoto, N. Masuda, **R. Hirayama**, H. Nakayama, T. Kakue, T. Shimobaba, and T. Ito, ‘Special-purpose computer for electroholography in embedded systems’, *OSA Continuum* 2(4), 1166–1173 (2019).
14. A. Shiraki, D. Matsumoto, **R. Hirayama**, H. Nakayama, T. Kakue, T. Shimobaba, and T. Ito, ‘Improvement of an algorithm for displaying multiple images in one space’, *Applied Optics* 58(5), A1–A6 (2019).
15. T. Nishitsuji, Y. Yamamoto, T. Sugie, T. Akamatsu, **R. Hirayama**, H. Nakayama, T. Kakue, T. Shimobaba, and T. Ito, ‘Special-purpose computer HORN-8 for phase-type electro-holography’, *Optical Express* 26(20), 26722 (2018).
16. T. Kakue, Y. Wagatsuma, S. Yamada, Y. Endo, Y. Nagahama, **R. Hirayama**, T. Shimobaba, and T. Ito, ‘Review of real-time reconstruction techniques for aerial-projection holographic displays’, *Optical Engineering* 57(6), 061621 (2018).
17. T. Sugie, T. Akamatsu, T. Nishitsuji, **R. Hirayama**, N. Masuda, H. Nakayama, Y. Ichihashi, A. Shiraki, M. Oikawa, N. Takada, Y. Endo, T. Kakue, T. Shimobaba, and T. Ito, ‘High-performance parallel computing for next-generation holographic imaging’, *Nature Electronics* 1, 254–259 (2018).
18. T. Shimobaba, Y. Endo, T. Nishitsuji, T. Takahashi, Y. Nagahama, S. Hasegawa, M. Sano, **R. Hirayama**, T. Kakue, A. Shiraki, and T. Ito, ‘Computational ghost imaging using deep learning’, *Optics Communications* 413, 147–151 (2018).
19. T. Shimobaba, K. Matsushima, T. Takahashi, Y. Nagahama, S. Hasegawa, M. Sano, **R. Hirayama**, T. Kakue, and T. Ito, ‘Fast, large-scale hologram calculation in wavelet domain’, *Optics Communications* 412, 80–84 (2018).
20. A. Shiraki, M. Ikeda, H. Nakayama, **R. Hirayama**, T. Kakue, T. Shimobaba, and T. Ito, ‘Efficient method for fabricating a directional volumetric display using strings displaying multiple images’, *Applied Optics* 57(1), A33–A38 (2018).
21. T. Shimobaba, N. Kuwata, M. Honma, T. Takahashi, Y. Nagahama, M. Sano, S. Hasegawa, **R. Hirayama**, T. Kakue, A. Shiraki, N. Takada, and T. Ito, ‘Convolutional neural network-based data page classification for holographic memory’, *Applied Optics* 56(26), 7327–7330 (2017).
22. **R. Hirayama**, A. Shiraki, H. Nakayama, T. Kakue, T. Shimobaba, and T. Ito, ‘Operating scheme of a light-emitting diode array for a volumetric display exhibiting multiple full-color dynamic images’, *Optical Engineering* 56(7), 073108 (2017).
23. **R. Hirayama**, T. Suzuki, T. Shimobaba, A. Shiraki, M. Naruse, H. Nakayama, T. Kakue, and T. Ito, ‘Inkjet printing-based volumetric display projecting multiple full-colour 2D patterns’, *Scientific Reports* 7, 46511 (2017).
24. T. Shimobaba, Y. Endo, **R. Hirayama**, Y. Nagahama, T. Takahashi, T. Nishitsuji, T. Kakue, A. Shiraki, N. Takada, N. Masuda, and T. Ito, ‘Autoencoder-based holographic image restoration’, *Applied Optics* 56(13), F27–F30 (2017).
25. T. Shimobaba, Y. Endo, **R. Hirayama**, D. Hiyama, Y. Nagahama, S. Hasegawa, M. Sano, T. Takahashi, T. Kakue, M. Oikawa, and T. Ito, ‘Holographic micro-information hiding’, *Applied Optics* 56(4), 833–837 (2017).
26. **R. Hirayama**, A. Shiraki, M. Naruse, S. Nakamura, H. Nakayama, T. Kakue, T. Shimobaba, and T. Ito, ‘Optical addressing of multi-colour photochromic material mixture for volumetric display’, *Scientific Reports* 6, 31543 (2016).
27. T. Shimobaba, M. Makowski, Y. Nagahama, Y. Endo, **R. Hirayama**, D. Hiyama, S. Hasegawa, M. Sano, T. Kakue, M. Oikawa, T. Sugie, N. Takada, and T. Ito, ‘Color computer-generated hologram generation using the random phase-free method and color space conversion’, *Applied Optics* 55(15), 4159–4165 (2016).
28. **R. Hirayama**, H. Nakayama, A. Shiraki, T. Kakue, T. Shimobaba, and T. Ito, ‘Image quality improvement for a 3D structure exhibiting multiple 2D patterns and its implementation’, *Optics Express* 24(7), 7319–7327 (2016).
29. T. Sanpei, T. Shimobaba, T. Kakue, Y. Endo, **R. Hirayama**, D. Hiyama, S. Hasegawa, Y. Nagahama, M. Sano, M. Oikawa, T. Sugie, and T. Ito, ‘Optical encryption for large-sized images’, *Optics Communications* 361, 138–142 (2016).
30. T. Shimobaba, T. Kakue, Y. Endo, **R. Hirayama**, D. Hiyama, S. Hasegawa, Y. Nagahama, M. Sano, M. Oikawa, T. Sugie, and T. Ito, ‘Improvement of the image quality of random phase-free holography using an iterative method’, *Optics Communications* 355, 596–601 (2015).
31. T. Shimobaba, T. Kakue, Y. Endo, **R. Hirayama**, D. Hiyama, S. Hasegawa, Y. Nagahama, M. Sano, M. Oikawa, T. Sugie, and T. Ito, ‘Random phase-free kinoform for large objects’, *Optics Express* 23(13), 17269–17274 (2015).
32. **R. Hirayama**, M. Naruse, H. Nakayama, N. Tate, A. Shiraki, T. Kakue, T. Shimobaba, M. Ohtsu, and T. Ito, ‘Design, implementation and characterization of a quantum-dot-based volumetric display’, *Scientific Reports* 5, 8472 (2015), *highlighted in Nature Japan*

33. D. Arai, T. Shimobaba, K. Murano, Y. Endo, **R. Hirayama**, D. Hiyama, T. Kakue, and T. Ito, 'Acceleration of computer-generated hologram using tilted wavefront recording plane method', *Optics Express* **23**(2), 1740–1747 (2015).
34. T. Shimobaba, M. Makowski, T. Kakue, N. Okada, Y. Endo, **R. Hirayama**, D. Hiyama, S. Hasegawa, Y. Nagahama, and T. Ito, 'Numerical investigation of lensless zoomable holographic projection to multiple tilted planes', *Optics Communications* **333**, 274–280 (2014).
35. T. Shimobaba, T. Kakue, N. Okada, Y. Endo, **R. Hirayama**, D. Hiyama, and T. Ito, 'Ptychography by changing the area of probe light and scaled ptychography', *Optics Communications* **331**, 189–193 (2014).
36. T. Shimobaba, T. Kakue, M. Oikawa, N. Takada, N. Okada, Y. Endo, **R. Hirayama**, and T. Ito, 'Calculation reduction method for color computer-generated hologram using color space conversion', *Optical Engineering*, **53**(2), 024108 (2014).
37. T. Shimobaba, T. Kakue, M. Oikawa, N. Okada, Y. Endo, **R. Hirayama**, N. Masuda, and T. Ito, 'Non-uniform sampled scalar diffraction calculation using non-uniform fast Fourier transform', *Optics Letters* **38**(23), 5130–5133 (2013).
38. T. Shimobaba, M. Makowski, T. Kakue, M. Oikawa, N. Okada, Y. Endo, **R. Hirayama**, N. Masuda, and T. Ito, 'Lensless zoomable holographic projection using scaled Fresnel diffraction', *Optics Express* **21**(21), 25285–25290 (2013).
39. T. Shimobaba, H. Yamanashi, T. Kakue, M. Oikawa, N. Okada, Y. Endo, **R. Hirayama**, and T. Ito, 'Inline digital holographic microscopy using a consumer scanner', *Scientific Reports* **3**, 2664 (2013).
40. H. Nakayama, A. Shiraki, **R. Hirayama**, N. Masuda, T. Shimobaba, and T. Ito, 'Three-dimensional volume containing multiple two-dimensional information patterns', *Scientific Reports* **3**, 1931 (2013).

Conference Proceedings

1. J. Mukherjee, Z. Shen, G. Christopoulos, S. Subramanian, and **R. Hirayama**, 'To BEM or not to BEM?: Does Modelling Sound Scattering Improve Ultrasonic Mid-Air Haptics?', CHI 2025 (Late Breaking Work), Yokohama, Japan (04/2025).
2. G. Christopoulos, L. Gao, D. M. Plasencia, M. Betcke, **R. Hirayama**, and S. Subramanian, 'Temporal acoustic point holography', SIGGRAPH 2024 (Conference Papers), Denver, USA (07/2024).
3. L. Gao, G. Christopoulos, P. Mittal, R. Hirayama, and S. Subramanian, 'StableLev: Data-driven stability enhancement for multi-particle acoustic levitation', CHI 2024 (Paper), Hawaii, USA (05/2024).
4. L. Gao, P. Irani, S. Subramanian, G. Prabhakar, D. M. Plasencia, and **R. Hirayama**, 'DataLev: Mid-air data physicalisation using acoustic levitation', CHI 2023, Hamburg, Germany (04/2023).
5. **(invited) R. Hirayama**, G. Christopoulos, D. M. Plasencia, and S. Subramanian, 'Acoustic mid-air displays in the presence of sound-scattering objects', LDC 2023, Yokohama, Japan (04/2023).
6. L. Gao, J. Hardwick, D. M. Plasencia, S. Subramanian, and **R. Hirayama**, 'DataLev: Acoustophoretic data physicalisation', UIST 2022 (Demonstration), Bend, USA (10/2022).
7. S. Bansal, **R. Hirayama**, and S. Subramanian, 'Fluid-acoustics: Dynamic multi-slit metamaterial tuned using liquid droplets', Acoustofluidics 2022, Glasgow, Scotland (10/2022).
8. **(invited) R. Hirayama**, 'Multimodal 3D display using ultrasound', 3DSA 2021, Online (12/2021).
9. **(invited) R. Hirayama**, D. M. Plasencia, R. Montano, and S. Subramanian, 'Paradigm shift from high-speed single-point to high-speed multipoint algorithms for multimodal acoustic displays', IDW 2021, Online (12/2020).
10. **(invited) R. Hirayama**, D. M. Plasencia, N. Masuda, and S. Subramanian, 'Multimodal acoustic trapping display', IDW 2020, Online (12/2020).
11. **(invited) R. Hirayama**, D. M. Plasencia, N. Masuda, and S. Subramanian, 'Acoustic levitation for multimodal volumetric display', SPIE Optics + Photonics 2020 (Proc. SPIE 11463), Optical Trapping and Optical Micromanipulation XVII; 114630Q, Online (08/2020).
12. D. M. Plasencia, **R. Hirayama**, R. Montano, and S. Subramanian, 'GS-PAT: High-speed multi-point sound-fields for phased arrays of transducers', SIGGRAPH 2020 Technical Papers, Online (08/2020).
13. D. Matsumoto, **R. Hirayama**, N. Hoshikawa, H. Nakayama, T. Shimobaba, T. Ito, and A. Shiraki, 'GPU acceleration of algorithm to design directional volumetric display for real-time', IDW 2019 / 3DSA 2019, Sapporo, Japan (11/2019).
14. T. Murase, **R. Hirayama**, N. Hoshikawa, H. Nakayama, T. Shimobaba, T. Ito, and A. Shiraki, 'Development of volumetric display capable of transmitting information in different languages using language identification', IDW 2019 / 3DSA 2019, Sapporo, Japan (11/2019).
15. M. Baba, **R. Hirayama**, N. Hoshikawa, H. Nakayama, T. Shimobaba, T. Ito, and A. Shiraki, 'Implemented of images and sounds person tracking system using directional volumetric display', IDW 2019 / 3DSA 2019, Sapporo, Japan (11/2019).
16. D. Matsumoto, T. Murase, **R. Hirayama**, H. Nakayama, T. Shimobaba T. Ito, and A. Shiraki, 'Subjective image quality evaluation to compare algorithms for designing a directional volumetric display', IDW 2018, Nagoya, Japan (12/2018).
17. T. Murase, D. Matsumoto, **R. Hirayama**, H. Nakayama, T. Shimobaba, T. Ito, and A. Shiraki, 'Image quality improvement for 3D structure exhibiting multiple 2D patterns using convolutional neural networks', IDW 2018, Nagoya, Japan (12/2018).
18. D. Matsumoto, A. Shiraki, **R. Hirayama**, H. Nakayama, T. Kakue, T. Shimobaba, and T. Ito, 'Improvement of the algorithm for designing a 3D object exhibiting multiple 2D images', 3DSA 2018, Taipei, Taiwan (08/2018).
19. M. Ikeda, A. Shiraki, **R. Hirayama**, T. Kakue, T. Shimobaba, and T. Ito, 'Simulation of the projection mapping to a directional volumetric display', IDW 2017, Sendai, Japan (12/2017).
20. **R. Hirayama**, H. Nakayama, A. Shiraki, T. Kakue, T. Shimobaba, and T. Ito, 'Controllable color particles in a 3D crystal projecting multiple dynamic full-color images', ACM SIGGRAPH 2017 Posters, 73, Los Angeles, USA (07/2017).
21. **R. Hirayama**, T. Suzuki, T. Shimobaba, A. Shiraki, M. Naruse, H. Nakayama, T. Kakue, and T. Ito, 'Inkjet-printed 3D structure projecting multiple full-color images', OPIC IP2017, IP-20AM-1-5, Yokohama, Japan (04/2017).

22. F. Kawashima, **R. Hirayama**, A. Shiraki, H. Nakayama, T. Kakue, T. Shimobaba, and T. Ito ‘Gradation expression by overlap of voxels in volumetric display composed of photochromic materials’, IDW / AD 2016, 3DSAp2/3Dp2-1, Fukuoka, Japan (12/2016).
23. **R. Hirayama**, A. Shiraki, H. Nakayama, T. Kakue, T. Shimobaba, and T. Ito, ‘3-D crystal with a curved surface projecting multiple 2-D images’, ACM SIGGRAPH Asia 2016 Posters, 41, Macao, China (12/2016).
24. **R. Hirayama**, A. Shiraki, H. Nakayama, T. Kakue, T. Shimobaba, and T. Ito, ‘Refraction-compensating algorithm for a 3D glass structure exhibiting multiple 2D images’, FiO / LS 2016, JTh2A-68, Rochester, USA (10/2016).
25. M. Oikawa, D. Hiyama, **R. Hirayama**, S. Hasegawa, Y. Endo, T. Sugie, N. Tsumura, M. Kuroshima, M. Maki, G. Okada, C. Lei, Y. Ozeki, K. Goda, and T. Shimobaba, ‘A computational approach to real-time image processing for serial time-encoded amplified microscopy’, SPIE Photonics West BIOS 2016 (Proc. SPIE 9720), 97200E, San Francisco USA (03/2016).
26. (invited) A. Shiraki, H. Nakayama, **R. Hirayama**, T. Kakue, T. Shimobaba, and T. Ito, ‘Volumetric display containing multiple two dimensional information patterns’, IDW 2015, PRJ1-1, Otsu, Japan (12/2015).
27. **R. Hirayama**, A. Shiraki, H. Nakayama, T. Kakue, T. Shimobaba, and T. Ito, ‘3-D crystal exhibiting multiple 2-D images with directivity’, ACM SIGGRAPH Asia 2015 Posters, 1, Kobe, Japan (11/2015).
28. (invited) **R. Hirayama**, A. Shiraki, H. Nakayama, T. Kakue, T. Shimobaba, and T. Ito, ‘3-D crystal exhibiting multiple 2-D images with directivity’, VRCAI 2015, 33, Kobe, Japan (10/2015).
29. **R. Hirayama**, A. Shiraki, M. Naruse, H. Nakayama, N. Tate, T. Kakue, T. Shimobaba, and T. Ito, ‘Optically controlled quantum-dot-based volumetric display exhibiting multiple patterns’, JSAP-OSA Joint Symposia 2015, 15p-2F-10, Nagoya, Japan (09/2015).
30. (invited) **R. Hirayama**, M. Naruse, H. Nakayama, A. Shiraki, T. Kakue, T. Shimobaba, and T. Ito, ‘Optically controlled volumetric display exhibiting multiple two-dimensional patterns’, CC3DMR 2015, 340–341, Busan, South Korea (06/2015).
31. **R. Hirayama**, H. Nakayama, A. Shiraki, T. Kakue, T. Shimobaba, and T. Ito, ‘Development of volumetric display based on multi-bit color LED’, APCCAS 2014, 547–550, Okinawa, Japan (11/2014).
32. **R. Hirayama**, H. Nakayama, A. Shiraki, T. Kakue, T. Shimobaba, and T. Ito, ‘Volumetric display containing multiple two-dimensional color motion pictures’, SPIE DSS 2014 (Proc. SPIE 9117), 911717, Baltimore, USA (05/2014).
33. (invited) T. Kakue, N. Masuda, Y. Endo, **R. Hirayama**, N. Okada, T. Shimobaba, and T. Ito, ‘Special-purpose computer for real-time reconstruction of holographic motion picture’, OIT 2013 (Proc. SPIE 9042), 90420B, Beijing, China (11/2013).
34. **R. Hirayama**, R. Omura, Y. Kobayashi, A. Shiraki, H. Nakayama, T. Kakue, N. Masuda, T. Shimobaba, and T. Ito, ‘Development of a digitized volumetric display containing multiple two-dimensional patterns’, 3DSA 2013, P7-2, Osaka, Japan (06/2013).
35. **R. Hirayama**, H. Ando, A. Shiraki, H. Nakayama, T. Kakue, N. Masuda, T. Shimobaba, and T. Ito, ‘Image-quality improvement of multiple two-dimensional patterns contained in three-dimensional volume’, 3DSA 2013, S11-1, Osaka, Japan (06/2013).
36. **R. Hirayama**, T. Shimobaba, H. Nakayama, A. Shiraki, T. Kakue, N. Masuda, and T. Ito, ‘Optical encryption using three-dimensional volume containing multiple two-dimensional information patterns’, DHIP 2012, C015, Tokushima, Japan (09/2012).
37. **R. Hirayama**, T. Shimobaba, H. Nakayama, A. Shiraki, T. Kakue, N. Masuda, and T. Ito, “Optical encryption using three-dimensional volume containing multiple two-dimensional information patterns,” DHIP 2012, C015, Tokushima, Japan (2012.09).

Other Talks

1. **R. Hirayama**, ‘Acoustic Contactless Food Delivery and Processing’, Keynote Speech at Culinary Action! On The Road (02/2025).
2. **R. Hirayama**, ‘Acoustic Levitation for Displays and Fabrication’, Talk at the High-Beams Seminar series (11/2024).
3. **R. Hirayama**, D. M. Plasencia, and S. Subramanian, ‘Wingardium leviosa: The ability to make objects levitate’, The annual Royal Academy of Engineering Research Forum (11/2022).
4. **R. Hirayama**, ‘A Multimodal Volumetric Display Using Acoustic Holography’, OSA Webinar: Novel Displays for Future Vision Science (2021).

Other Articles

1. **R. Hirayama** and S. Subramanian, ‘Magical multi-modal displays using acoustophoresis’, XRDS: Crossroads, The ACM Magazine for Students **29**(1), 54–58 (2022).
2. **R. Hirayama**, A. Shiraki, T. Kakue, T. Shimobaba, and T. Ito, ‘Optical addressing method for full-color 3D display’, SPIE Newsroom (2016).

Patents

1. **R. Hirayama**, G. Christopoulos, D. M. Plasencia, and S. Subramanian, ‘High-speed acoustic holography with arbitrary scattering objects’, UK Patent Application No. GB2207885.1 (2022).
2. **R. Hirayama**, D. M. Plasencia, and S. Subramanian, ‘Multimodal acoustic trap display (MAT)’, UK Patent Application, No. GB1914174.6 (2019), Publication No. WO2021064403A1 (2021).
3. T. Ito, **R. Hirayama**, A. Shiraki, T. Kakue, and T. Shimobaba, ‘3D display device and its display method’, Japanese Unexamined Patent Publication No. 2017-191255 (2017).

4. T. Shimobaba, **R. Hirayama**, T. Sugie, Y. Endo, M. Oikawa, K. Goda and M. Ugawa, ‘Signal data processing device’, Japanese Unexamined Patent Publication No. 2017-134044 (2017).
5. T. Ito, M. Naruse, **R. Hirayama**, H. Nakayama, and T. Kakue, ‘Quantum-dot display and its display method’, Japanese Unexamined Patent Publication No. 2015-165611 (2015).

Grants and Awards

1. **Higher Education Innovation Fund (HEIF)**, No. 574843, ‘Acoustophoretic manufacturing’, UCL, £5K, CoI (12/2022–11/2021).
2. **UKRI guarantee funding for ERC Starting Grants**, No. EP/X019519/1, ‘FabDisp: Acoustic holography for multimodal 3D display and fabrication’, UKRI, £1.5M, PI (09/2022–09/2027).
3. **Grant-in-Aid for JSPS Fellows**, No. 18J01002, Japan Society for the Promotion of Science, ¥1.3M (~£9.7K) + salary (~£24.2K) (04/2018–12/2018).
4. **Inoue Research Award for Young Scientists**, Inoue Foundation for Science, ¥500K (~£3.7K) (02/2018).
5. **Young Researcher Award**, Kenjiro Takayanagi Foundation, ¥2M (~£14.9K) (01/2018).
6. **President Award for the Excellent Record**, Chiba University (03/2017).
7. **Dean Award for the Excellent Record**, Graduate School of Engineering, Chiba University (03/2017).
8. **GP Program to Support Sending Graduate Students Abroad**, Institute for Global Prominent Research, Chiba University, ¥136K (~£1.0K) (12/2017).
9. **Program to Support Sending Graduate Students Abroad**, Chiba University, ¥136K (~£1.0K) (10/2017).
10. **Grant-in-Aid for JSPS Fellows**, No. 16J30007, Japan Society for the Promotion of Science, ¥2.3M (~£17.1K) + salary (~£35.7K) (04/2016–03/2018).
11. **JSPS Ikushi Prize**, Japan Society for the Promotion of Science, ¥2M (~£14.9K) (03/2016).
12. **KONICA MINOLTA Science and Technology Foundation Award**, The Optical Society of Japan, ¥50K (~£0.4K) (06/2015).
13. **Scholarship Loan Forgiveness for Academic Excellence (Full Amount)**, Japan Student Services Organization (05/2015).
14. **Best Poster Award (FORUM 8 Award)**, Computer Graphic Arts Society (03/2015).
15. **Outstanding Paper Award for Young C&C Researchers**, NEC C&C Foundation, ¥200K (~£1.5K) (01/2015).
16. **Grant-in-Aid for JSPS Fellows**, No. 15J07684, Japan Society for the Promotion of Science, ¥1.2M (~£9.0K) + salary (~£17.9K) (04/2015–03/2016).
17. **Scholarship Loan Forgiveness for Academic Excellence (Full Amount)**, Japan Student Services Organization (05/2014).
18. **Program to Support Sending Graduate Students Abroad**, Chiba University, ¥150K (~£1.1K) (04/2014).
19. **Grants for Researchers Attending International Conferences**, NEC C&C Foundation, ¥200K (~£1.5K) (04/2014).
20. **President Award for the Excellent Record**, Chiba University (03/2014).
21. **Dean Award for the Excellent Record**, Graduate School of Engineering, Chiba University (03/2014).

Media

1. **BBC Earth Kids** (YouTube video), ‘Can sound kill you?’ (04/2022).
2. **New Scientist Video** (YouTube video), ‘Watch sonic tractor beams lift and manipulate objects with sound waves’ (09/2021).
3. **IEEE Signal Processing Magazine** (Technical magazine), ‘Three New Imaging Technologies That Are Worth a Look’ (09/2020).
4. **El Hormiguero 3.0** (Prime-time Spanish TV show), ‘LEVITACIÓN ACÚSTICA’ (02/2020).
5. **ITV News** (British TV show), ‘How holograms could be the future of medicine, advertising and entertainment’ (11/2019).
6. **BBC Radio** (British radio channel), ‘Science in Action’ (11/2019).
7. **The Guardian** (British Newspaper), ‘Hologram-like device animates objects using ultrasound waves’ (11/2019).
8. **Discover Magazine** (American science magazine), ‘With a Floating Bead, This Device Makes Truly 3D Holographs’ (11/2019).
9. **Scientific American** (American science magazine), ‘Hearing Is Seeing: Sound Waves Create a 3-D Display’ (11/2019).
10. **Science Magazine** (News site of AAAS/Science), ‘See the new Star Wars-like display that could ‘revolutionize’ virtual reality’ (11/2019).
11. **NBC News** (American news site), ‘With a single bead of plastic, scientists tease interactive 3D’ (11/2019).
12. **NewScientist** (British science magazine), ‘We can now make animated ‘sound holograms’ that you can touch’ (11/2019).
13. **日本経済新聞** (Japanese Newspaper), ‘立体映像とともに音・触感を再現 英大学が技術’ (11/2019).
14. **Dagens Nyheter** (Swedish Newspaper), ‘Ny teknik skapar 3D-bilder som låter och känns’ (11/2019).
15. **NRC Handelsblad** (Dutch Newspaper), ‘3D-beeld gevangen in geluidsgolven’ (11/2019).
16. **Neue Zürcher Zeitung** (Swiss Newspaper), ‘Fast ein richtiges Holodeck’ (11/2019).
17. **Cité des sciences et de l'industrie** (French museum website), ‘Du virtuel au bout des doigts!’ (11/2019).
18. **OSA Publishing, Image of the Week**, ‘Projection of multiple directional images on a volume structure with refractive surfaces’ (10/2019).

19. **OSA Publishing, Image of the Week**, 'Image quality improvement for a 3D structure exhibiting multiple 2D patterns and its implementation' (04/2016).
20. **科学新聞** (Japanese scientific newspaper), '育志賞 - 18人に' (03/2016).
21. **TBS** (Japanese TV show), '未来の起源' (09/2015).
22. **日経産業新聞** (Japanese newspaper), '紫外線当て3D表示—千葉大など装置考案' (04/2015).
23. **日刊工業新聞** (Japanese newspaper), '光を当てると各面に異なる画像を表示—千葉大, 新原理の立体ディスプレイ開発' (02/2015).
24. **テレビ東京** (Japanese TV show), 'ワールドビジネスサテライト (トレンドたまご)' (09/2013).
25. **日刊工業新聞** (Japanese newspaper), '見る位置で違う映像に—千葉大ディスプレイ開発' (09/2013).
26. **日刊工業新聞** (Japanese newspaper), '見る方向で違い—千葉大技術開発' (06/2011).

Membership

10/2016–present	Association for Computing Machinery (ACM)
08/2016–present	The Optical Society (OSA)

Skills

Programming

C, C++, C#, Python, CUDA, OpenCL, Matlab, VHDL, Verilog HDL, OpenGL, OpenCV, HTML/CSS

Others

Electronics (FPGA design and Microcontroller), Real-time system (GPU computing, Multi threads, and FPGA), Fabrication (3D printer, Laser cutter, and Luminescent materials), Projection mapping, 2D/3D design (Illustrator and Maya).